Final Project:

"Not every DHCP server is your friend"



Made By:

Ammar Yaser

Protecting Your Network: DHCP Spoofing, DHCP Snooping & Port Security

Ensuring the integrity and security of your network infrastructure is paramount in today's digital landscape. This presentation will explore critical Layer 2 security measures that safeguard your network against common threats like DHCP spoofing and unauthorized access.



The Impact of DHCP Spoofing Attacks

1

DHCP Starvation

Flooding DHCP requests with spoofed MACs exhausts IP pools, denying service to legitimate clients. This can bring your network to a halt.

2

Man-In-The-Middle (MITM)

Combines starvation and rogue DHCP to intercept sensitive data and harvest credentials. Imagine your login details being stolen in plain sight.

3

Real Risk

Unauthorized network access, data interception, and cascading security breaches can lead to significant financial and reputational damage for your organization.

What is DHCP Spoofing?

1

Rogue DHCP Servers: Attackers set up unauthorized DHCP servers to respond to client requests with malicious IP configurations.

2

Traffic Interception: This enables interception of traffic, man-in-the-middle attacks, and overall network disruption.

3

Example: A rogue DHCP server might list itself as the default gateway or DNS server, redirecting all client traffic through the attacker's machine for interception.



DHCP Snooping: Your First Line of Defense

DHCP Snooping is a crucial switch feature that actively monitors DHCP messages. It acts as a gatekeeper, allowing only trusted DHCP servers to respond to client requests. It also maintains a binding table, linking MAC addresses to IP addresses and ports, and proactively blocks DHCP responses from any rogue servers. Additionally, it limits DHCP request rates to prevent flooding attacks, securing your network from service disruptions.



How DHCP Snooping Works in Practice

01

Trusted Ports

Designate specific switch ports as "trusted." These are typically uplink ports connected directly to your legitimate DHCP servers, allowing all DHCP traffic to pass through.

02

Untrusted Ports

All other ports, especially those connecting to end-user devices, are considered "untrusted." Any DHCP replies originating from these untrusted ports are automatically blocked, neutralizing rogue server attempts.

03

Rate Limiting

Implement rate limiting on untrusted ports to prevent DHCP starvation attacks. This controls the number of DHCP requests a port can send, blocking floods of spoofed MAC addresses.

04

Example Configuration

Enable snooping on relevant VLANs (e.g., `ip dhcp snooping vlan 10,20`) and mark the server-facing port as trusted (e.g., `ip dhcp snooping trust`).

Port Security: Controlling Who Connects

Port security is a Layer 2 access control feature that restricts input to a port by limiting and identifying MAC addresses of allowed devices. It's a critical component in preventing unauthorized devices from connecting to your network.

Maximum MACs

Limits the number of MAC addresses that can be learned or configured on a switch port, preventing MAC address flooding attacks and controlling device density.

MAC Learning

Supports static (manually configured), dynamic (automatically learned), and sticky (dynamically learned and saved) MAC address configurations.

Violation Actions

When a violation occurs, the port can be shut down, traffic can be restricted, or packets from unauthorized MACs can be dropped while the port remains up.

Example: `switchport port-security maximum 3` on an access port allows only up to 3 unique MAC addresses to connect, effectively limiting the number of devices.

Combining DHCP Snooping & Port Security

DHCP Snooping

Focuses on validating DHCP messages and ensuring responses come only from authorized DHCP servers.

Port Security

Controls physical device connections by limiting the number of allowed MAC addresses per switch port.

Together, these two mechanisms create a robust defense, preventing both rogue DHCP servers from operating and deterring DHCP starvation attacks by limiting the number of devices per port.

⚠ **Important Caveat:** Some switches might allow DHCP bindings from spoofed MACs to persist even if a port security violation occurs. Careful configuration, regular monitoring, and awareness of platform-specific behaviors are essential for full protection.

Real-World Configuration Example

Device	Cisco 2960 Series Switch
DHCP Snooping	Enabled on VLANs 10, 20, 50, 100, 150. (`ip dhcp snooping vlan 10,20,50,100,150`)
Trusted Port	Uplink port configured as trusted for legitimate DHCP traffic. (`ip dhcp snooping trust`)
Access Ports	Configured with sticky MAC addresses and a maximum of 3 devices per port. (`switchport port-security mac-address sticky`, `switchport port-security maximum 3`)
Rate Limiting	DHCP requests limited to 10 packets/second on untrusted ports to prevent floods. (`ip dhcp snooping limit rate 10`)

Best Practices & Recommendations

- Always enable DHCP snooping on all VLANs with DHCP clients to ensure comprehensive coverage.
- Mark uplink ports as trusted; all other client-facing ports should be designated as untrusted.
- Configure port security with appropriate MAC limits and sticky addresses for enhanced control.
- Regularly monitor DHCP snooping binding tables and port security violations to detect anomalies.
- Combine with other Layer 2 protections like Dynamic ARP Inspection (DAI) and IP Source Guard for a robust defense.

Layer 2 Security is Essential

DHCP Spoofing represents a significant threat that exploits fundamental protocol weaknesses. However, with the strategic implementation of DHCP Snooping and Port Security, you gain powerful defenses.

DHCP Spoofing



A serious threat that can lead to data interception and network disruption.

DHCP Snooping



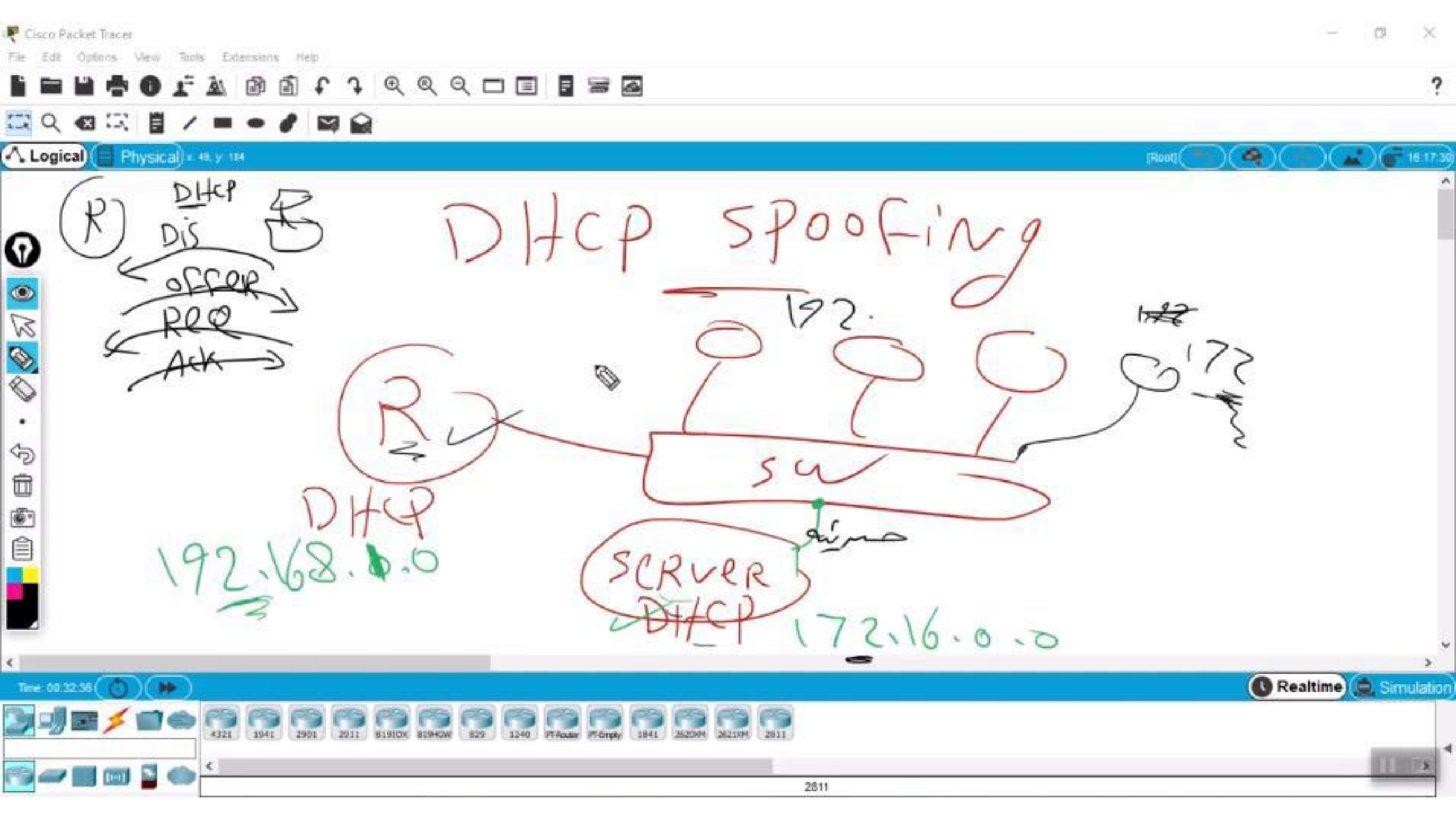
Your primary defense against rogue DHCP servers, ensuring trusted communication.

Port Security



Restricts access by limiting MAC addresses, preventing unauthorized connections and MAC spoofing.

Proper configuration, consistent monitoring, and a layered security approach are key to maintaining a secure and resilient network infrastructure. Protect your network today by mastering these essential Layer 2 defenses!

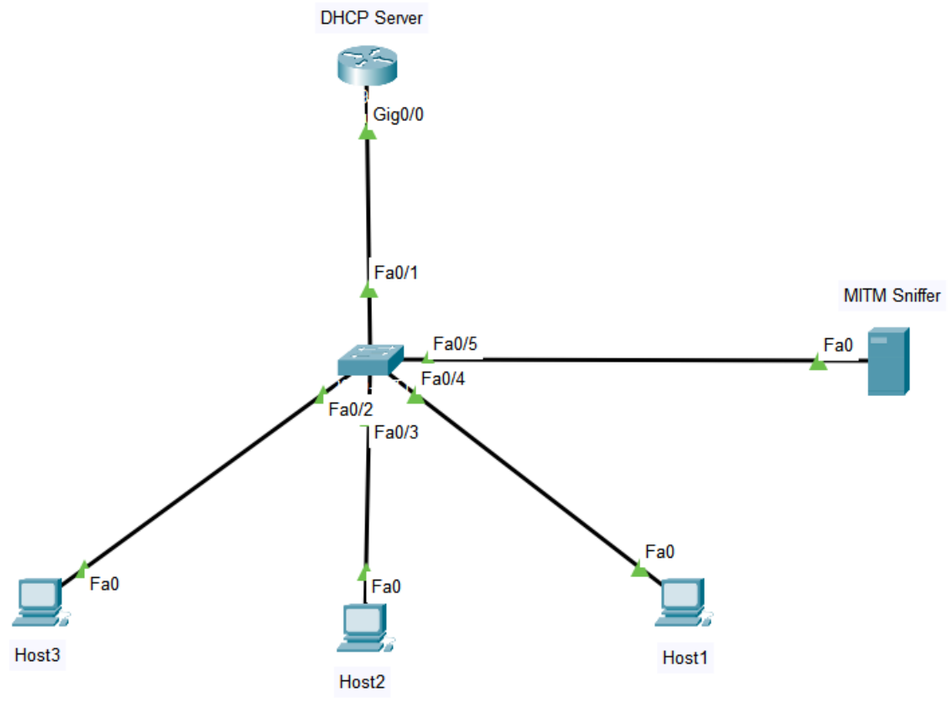

Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
Physical
DHCP
DIS
OFFER
REQ
ACK
DHCP SPOOFING
192.
172
R1
SW
SERVER
DHCP
DHCP
192.168.0.0
172.16.0.0
Realtime
Simulation
2811


Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
Physical
DHCP
DIS
OFFER
REQ
ACK
DHCP SPOOFING
SNOOPING
ATTACH
192.168.1.1
192.168.1.70
192.168.1.15
SW
DHCP
192.168.1.0
192.168.1.15
WIRESHARK
Realtime
Simulation
2811

Cisco Packet Tracer - C:\Users\DELL\OneDrive - Faculty Of Engineering (Zagazig University)\Desktop\Projects\Cisco Packet Tracer\saves\Ammar23 (DHCP Spoofing).pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1443, y: 82 Root 23:45:00

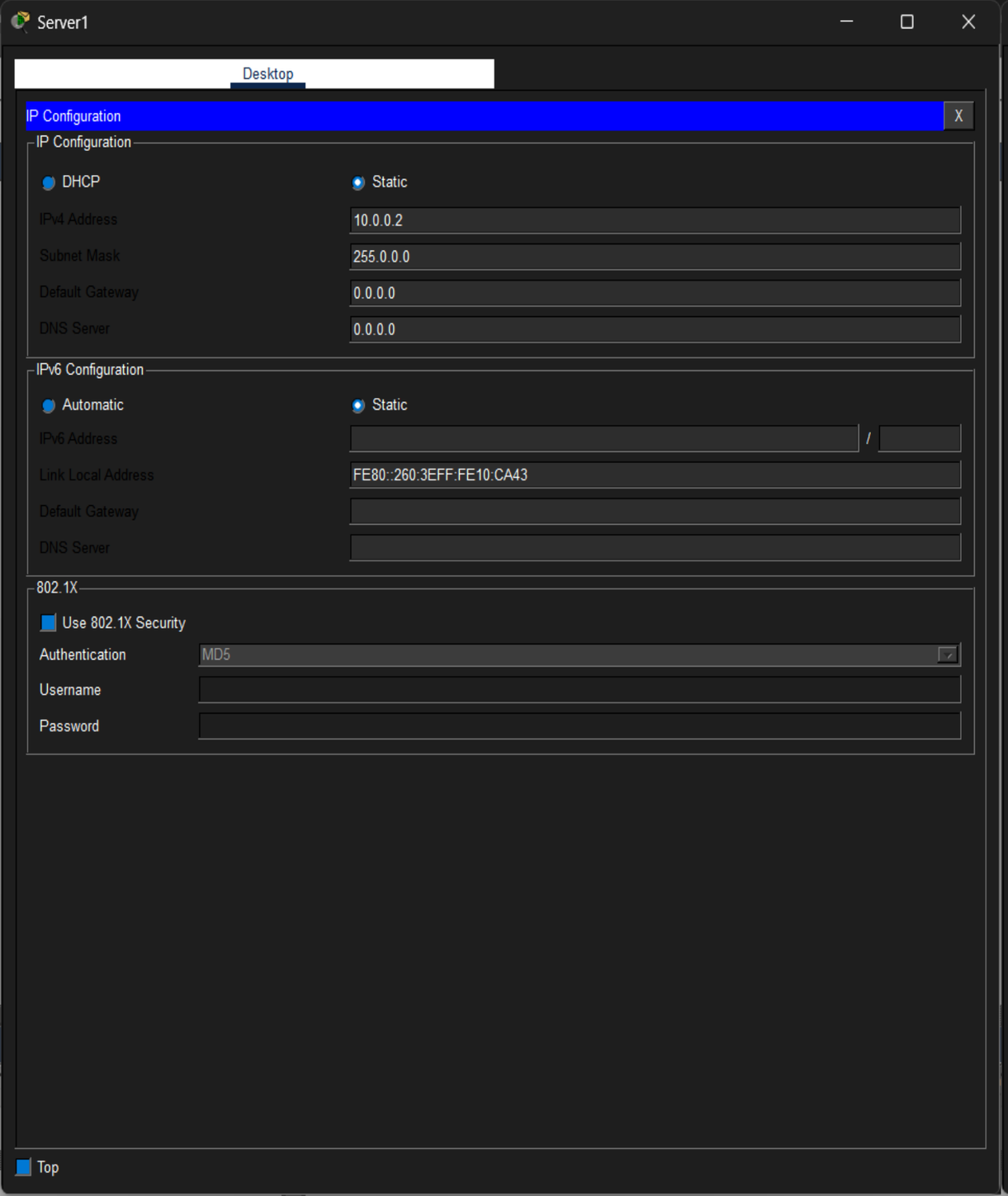


Time: 03:10:42

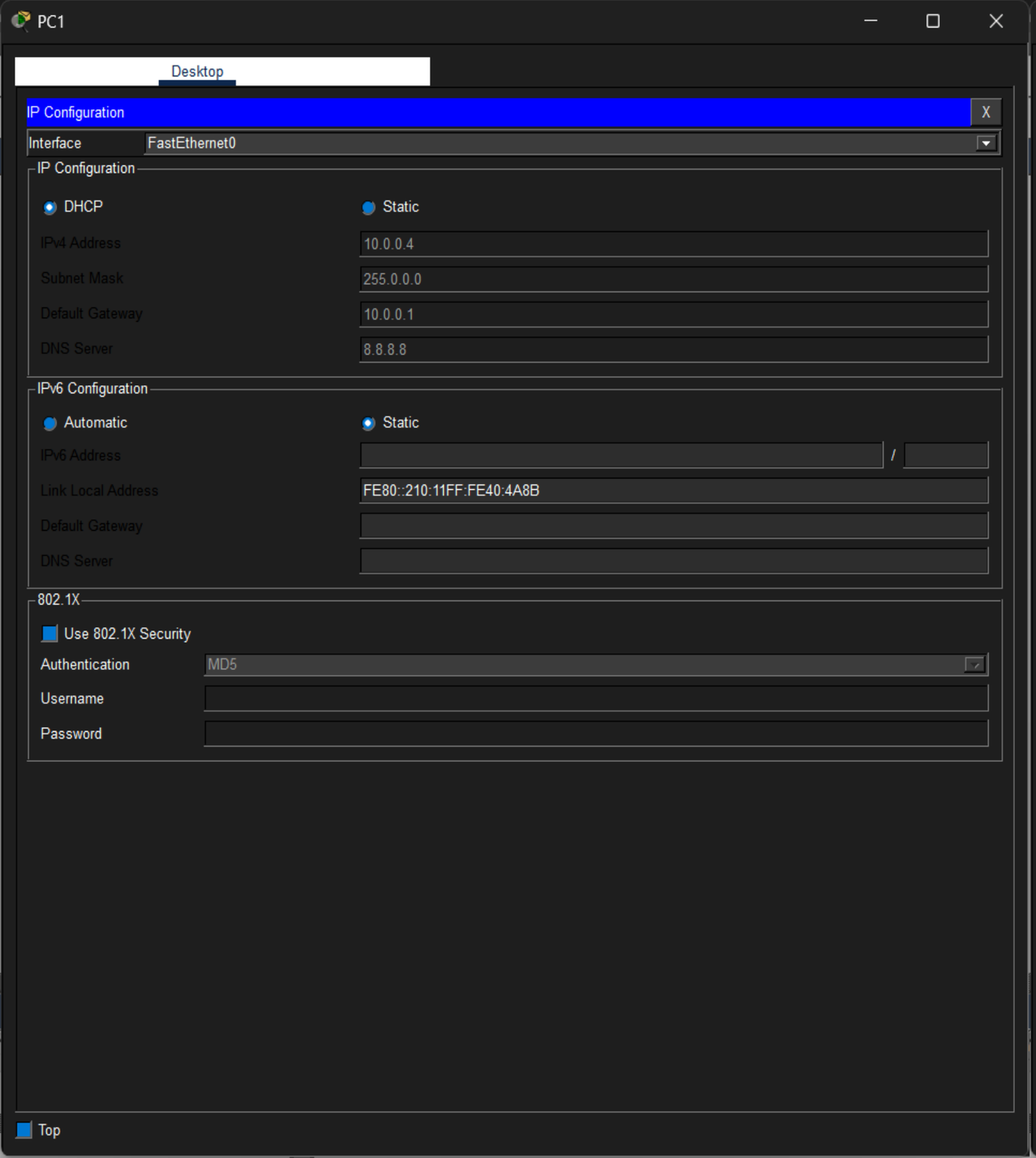
Realtime Simulation

Scenario 0 New Delete Toggle PDU List Window

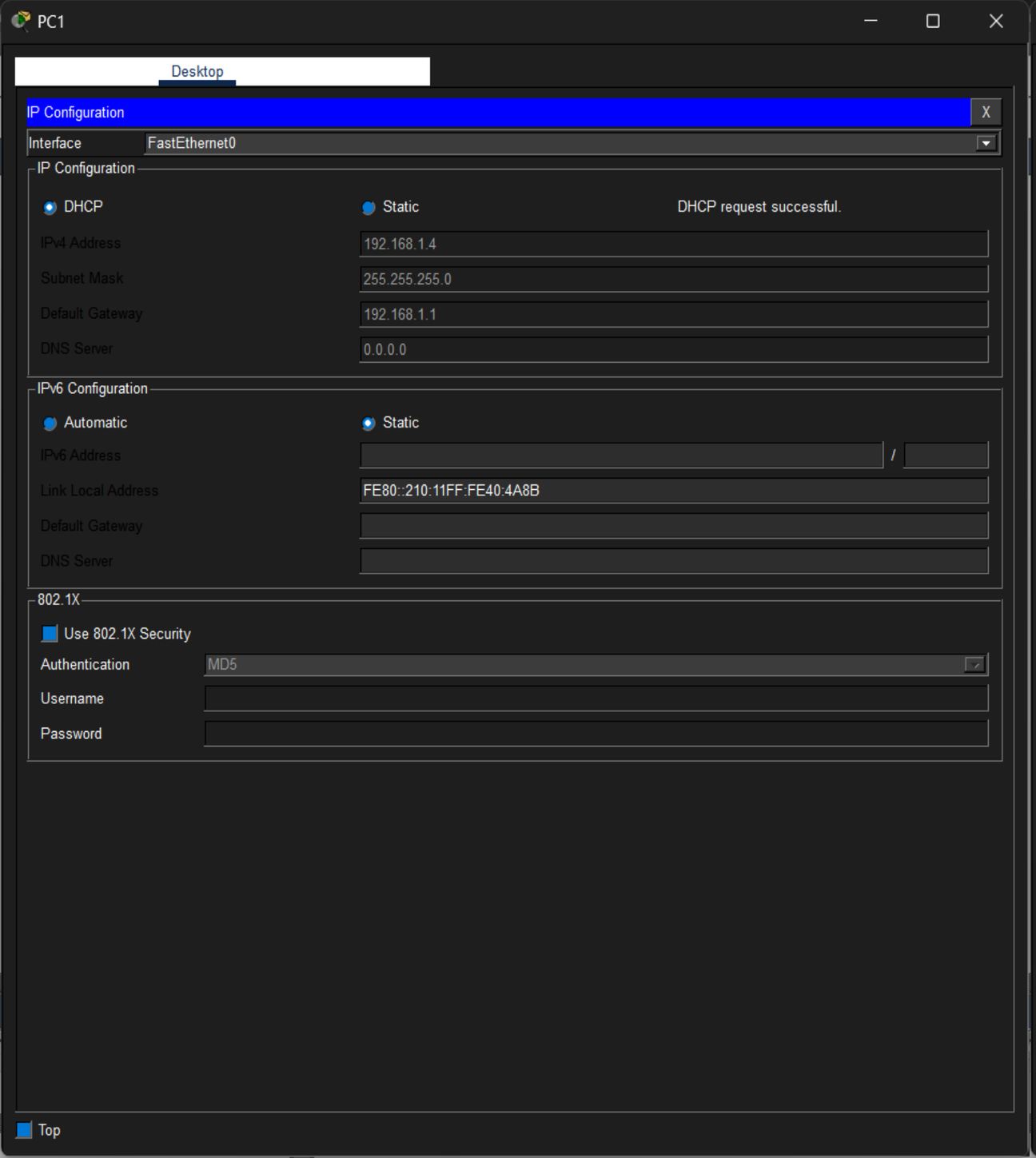
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete

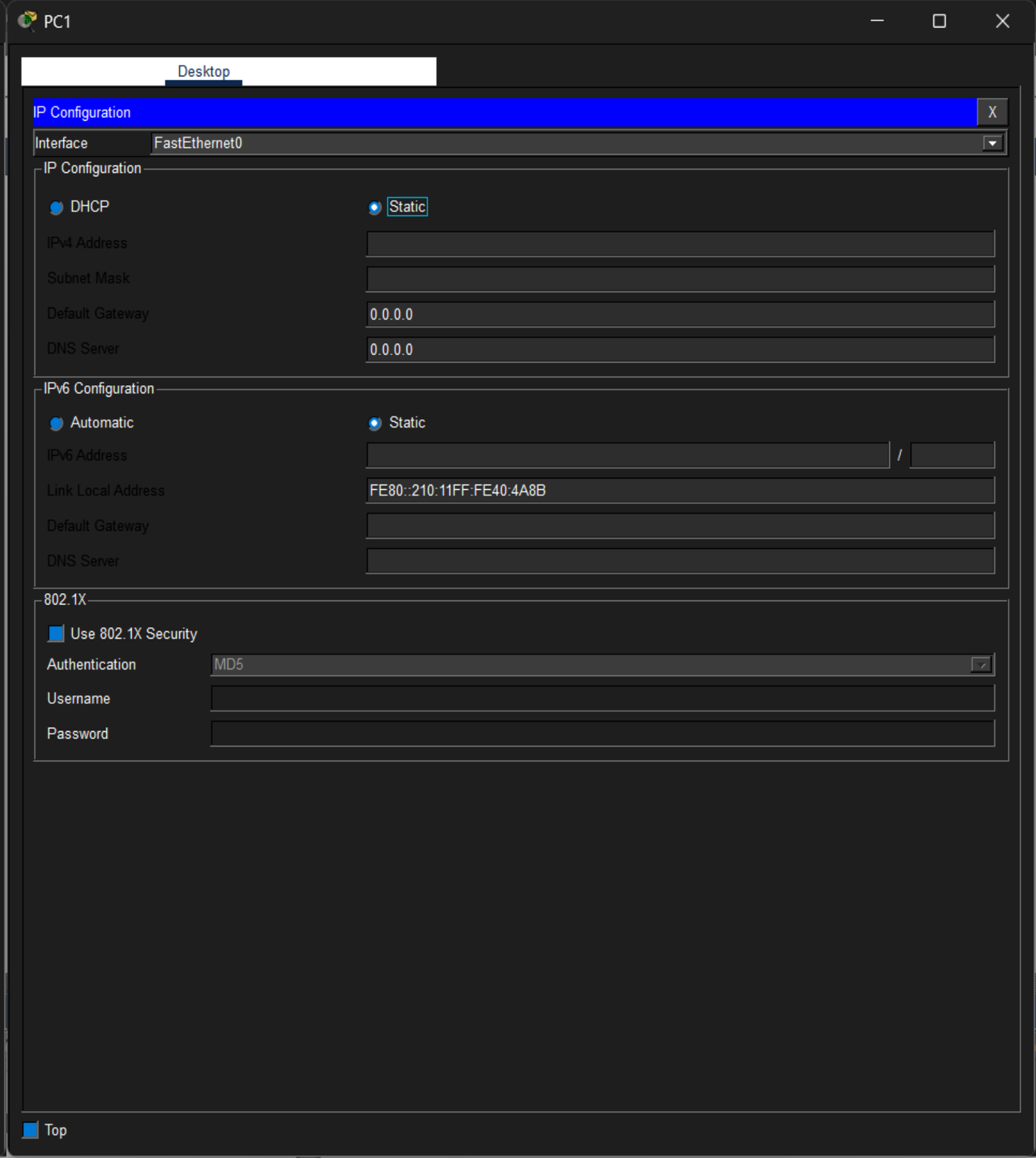

Server1
Desktop
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.2
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:3EFF:FE10:CA43
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Router5
CLI
Image text-base: 0x2100F918, data-base: 0x24729040
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: n
Press RETURN to get started!
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ho ammar
ammar(config)#ena se 123
ammar(config)#int g0/0
ammar(config-if)#ip ad 192.168.1.1 255.255.255.0
ammar(config-if)#n sh
ammar(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
ammar(config-if)#ex
ammar(config)#ip dhcp pool 1
ammar(dhcp-config)#net 192.168.1.0 255.255.255.0
ammar(dhcp-config)#def 192.168.1.1
ammar(dhcp-config)#do wr
ammar(dhcp-config)#
Copy
Paste
Top


PC1
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
10.0.0.4
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.1
DNS Server
8.8.8.8
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE40:4A8B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Server1
Services
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
FastEthernet0
serverPool
10.0.0.1
8.8.8.8
10
0
0
3
255
0
0
0
512
0.0.0.0
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
10.0.0.1
8.8.8.8
10.0.0.3
255.0.0.0
512
0.0.0.0
0.0.0.0
Top

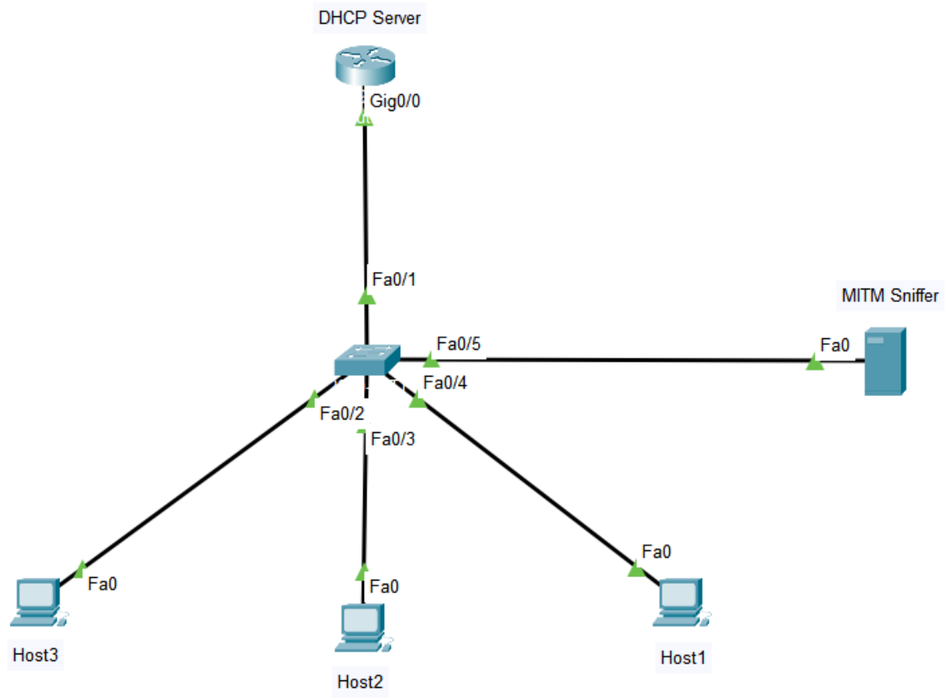

PC1
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE40:4A8B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC1
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE40:4A8B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

Cisco Packet Tracer - C:\Users\DELL\OneDrive - Faculty Of Engineering (Zagazig University)\Desktop\Projects\Cisco Packet Tracer\saves\Ammar24 (DHCP Snooping) with Port Security.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1641, y: 278 Root 00:19:30



Time: 00:00:39 Realtime Simulation

PT8200 4331 4321 IR8340 1941 2901 2911 819IOX 819HGW 829 1240 IR1101 PT-Router

Scenario 0 New Delete Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete

Switch0

CLI

```
Motherboard revision number     : B0
Model number                    : WS-C2960-24TT-L
System serial number            : FOC1010X104
Top Assembly Part Number        : 800-27221-02
Top Assembly Revision Number    : A0
Version ID                      : V02
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up


ammar>en
Password:
ammar#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ammar(config)#ip dh s
ammar(config)#no ip dh s i o
ammar(config)#ip dh s vl 1
ammar(config)#int ra f0/1 -4
ammar(config-if-range)#swi por mac st
ammar(config-if-range)#
```

Copy Paste

Top

Switch0

CLI

```
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Compiled Wed 26-Jun-13 02:49 by mnguyen



Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

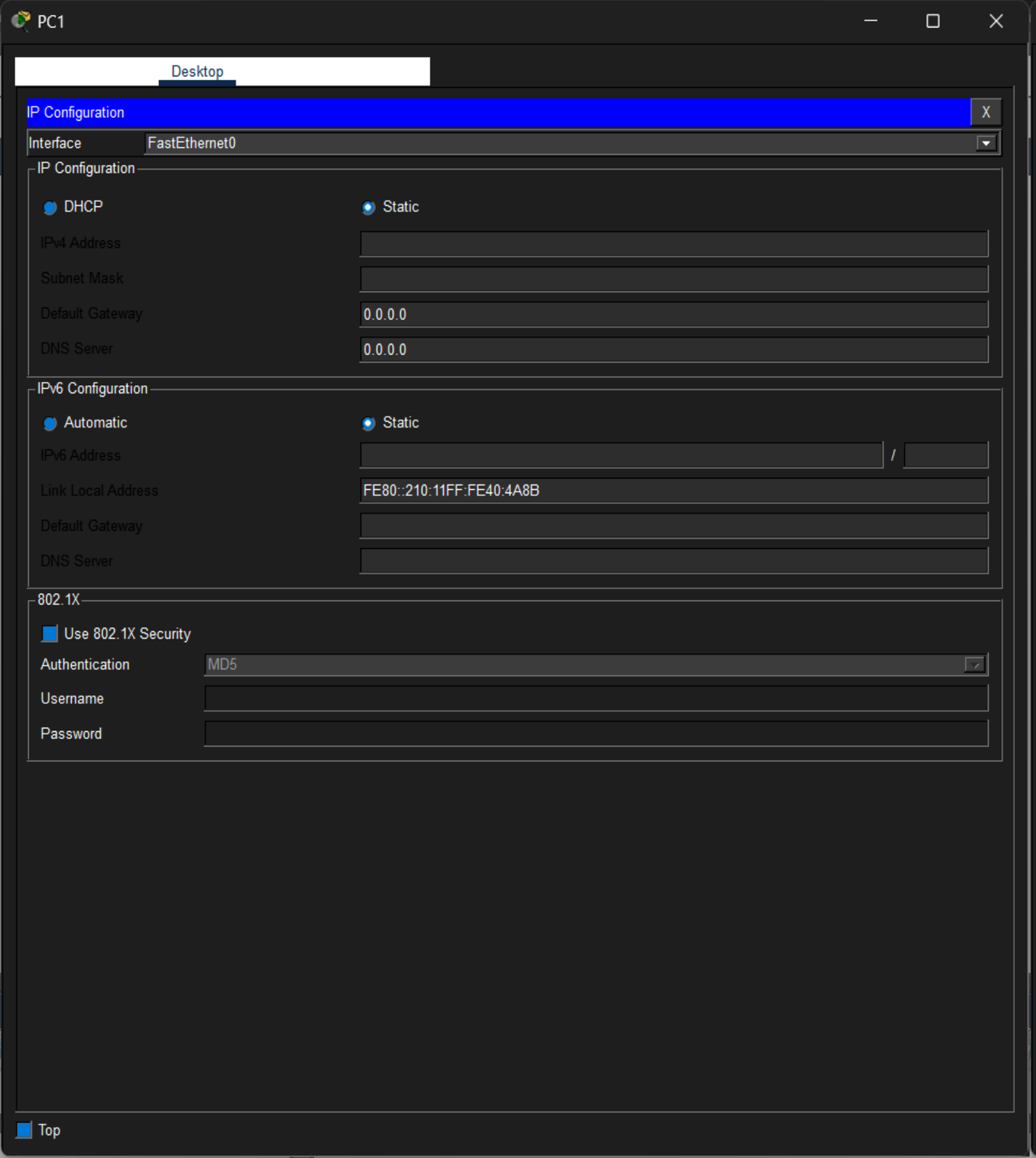
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

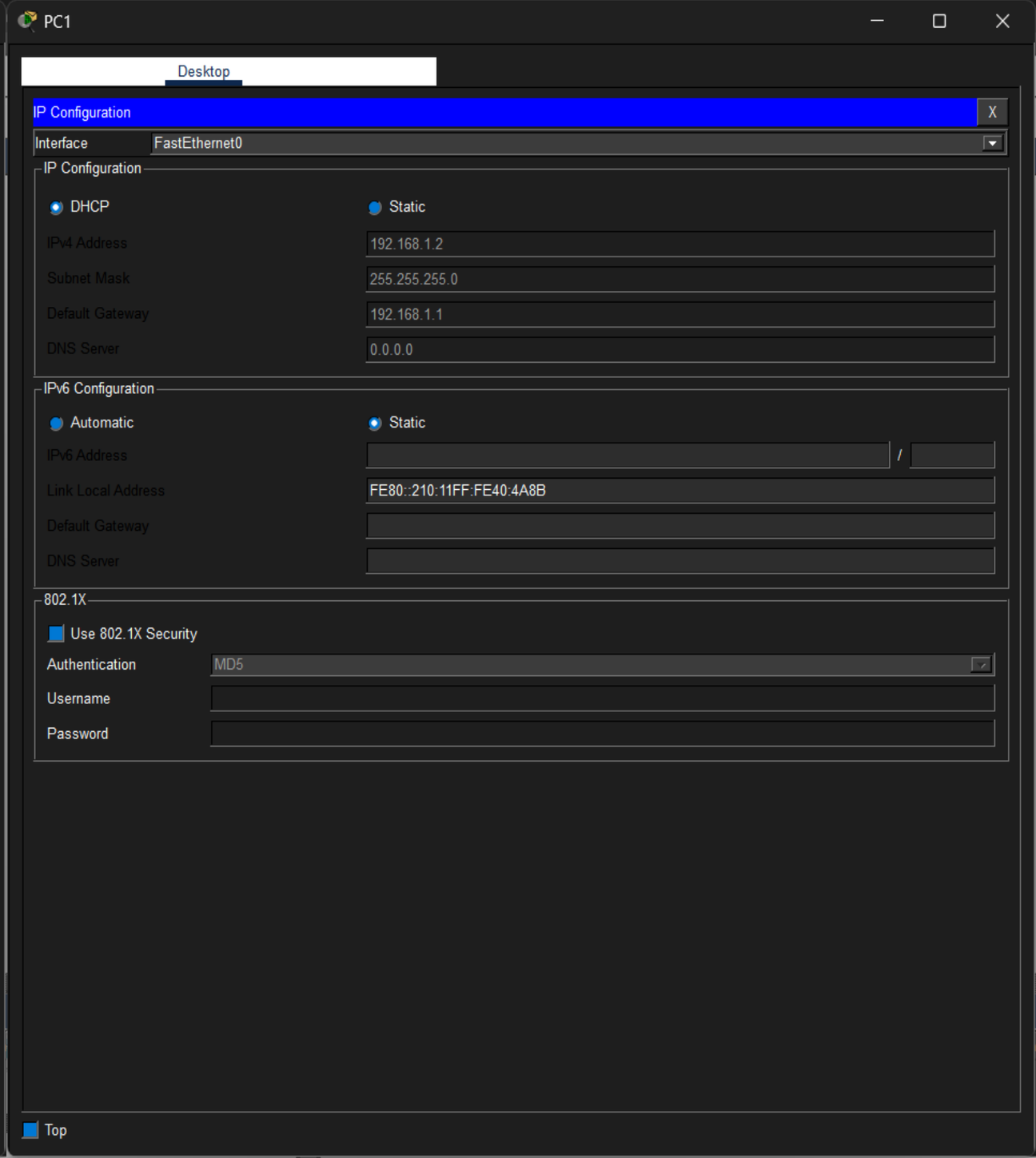
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

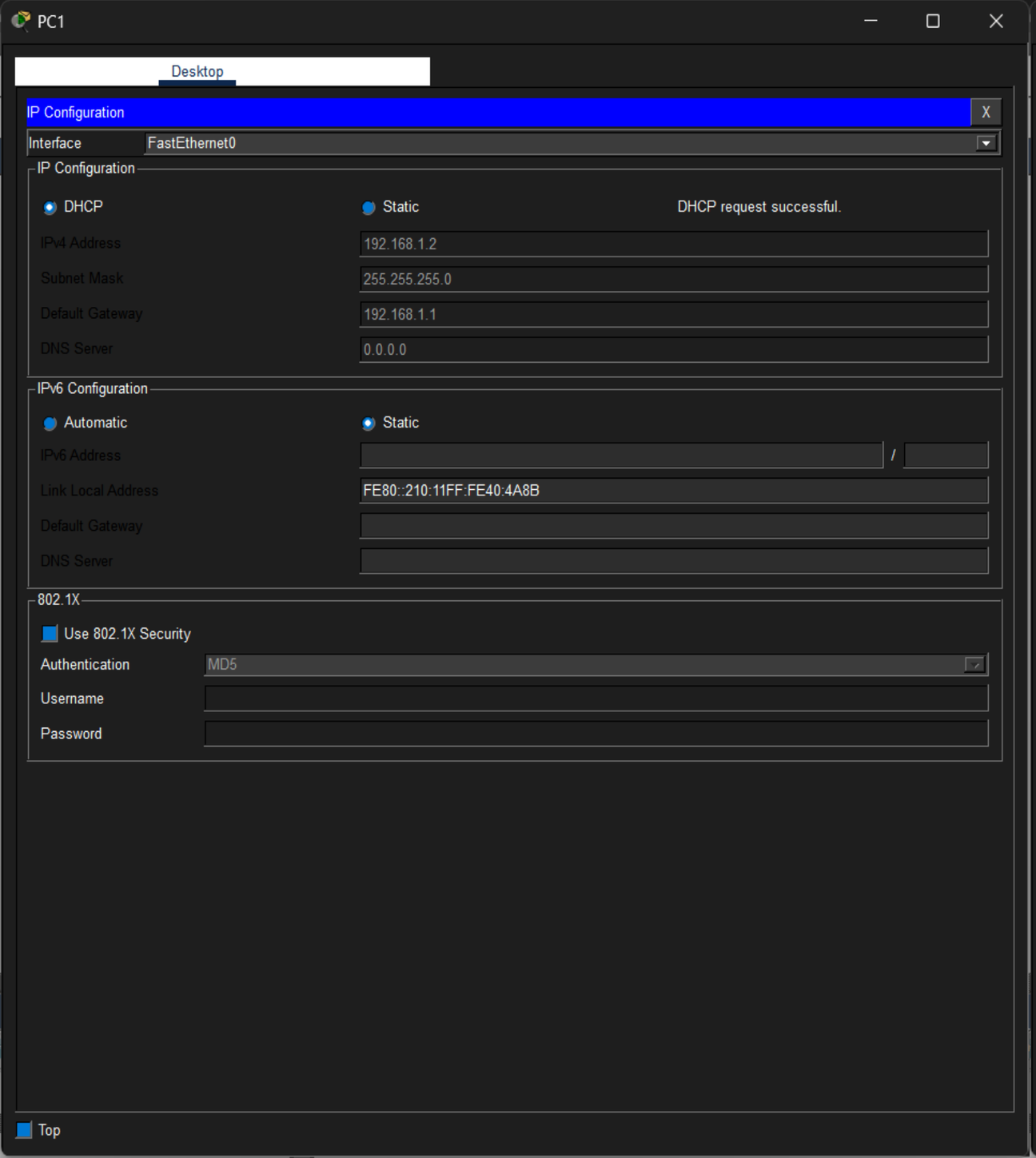

Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ho ammar
ammar(config)#ena se 123
ammar(config)#ip dh s
ammar(config)#no ip dh s i o
ammar(config)#ip dh s vlan 1
ammar(config)#int f0/1
ammar(config-if)#ip dh s t
ammar(config-if)#do wr
Building configuration...
[OK]
ammar(config-if)#do sh ip dh s
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
1
Insertion of option 82 is disabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                Trusted    Rate limit (pps)
-----------------------  -------    ----------------
FastEthernet0/1          yes        unlimited
ammar(config-if)#
```

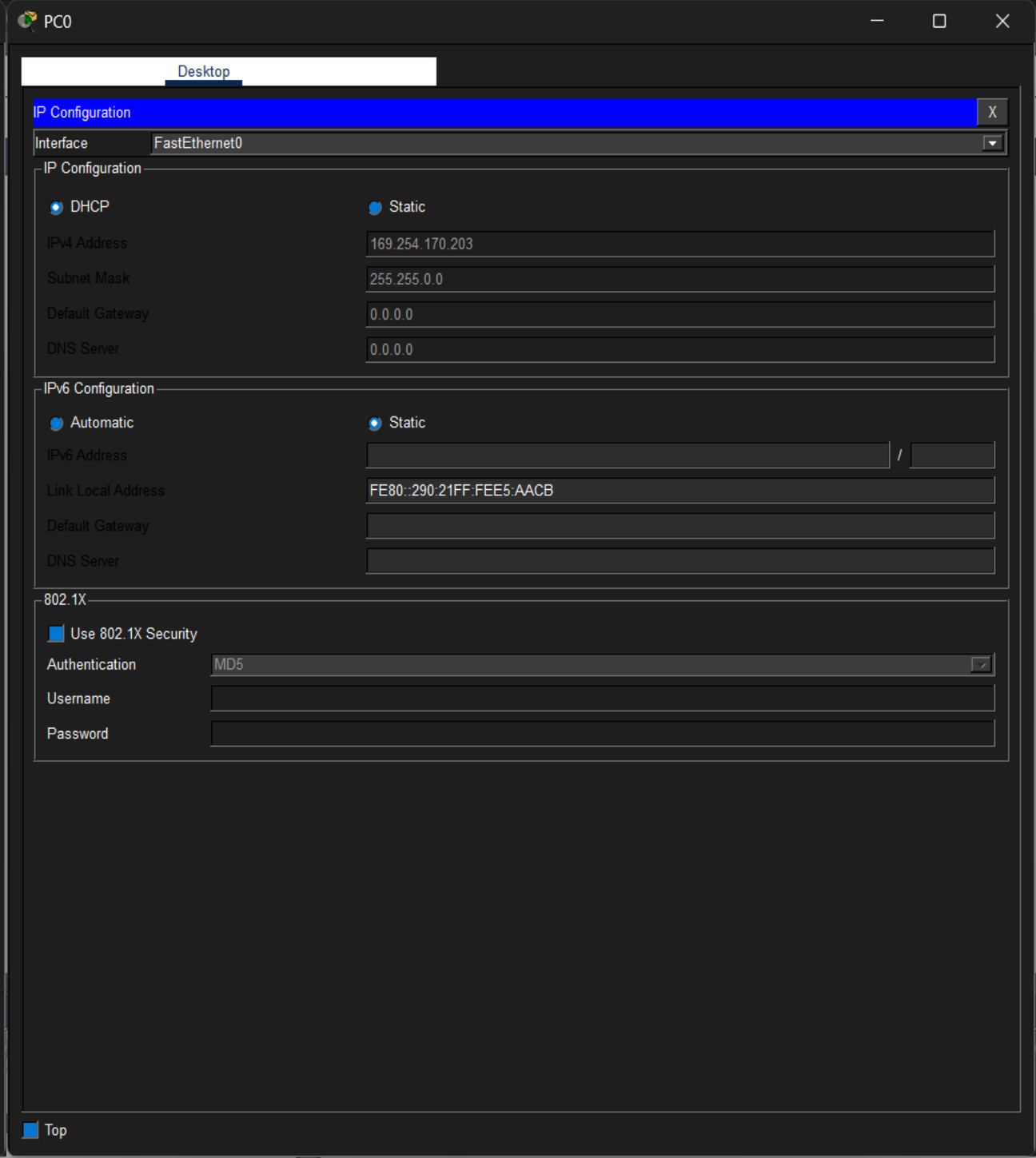
Copy Paste

Top


PC1
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
Subnet Mask
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE40:4A8B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC1
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE40:4A8B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

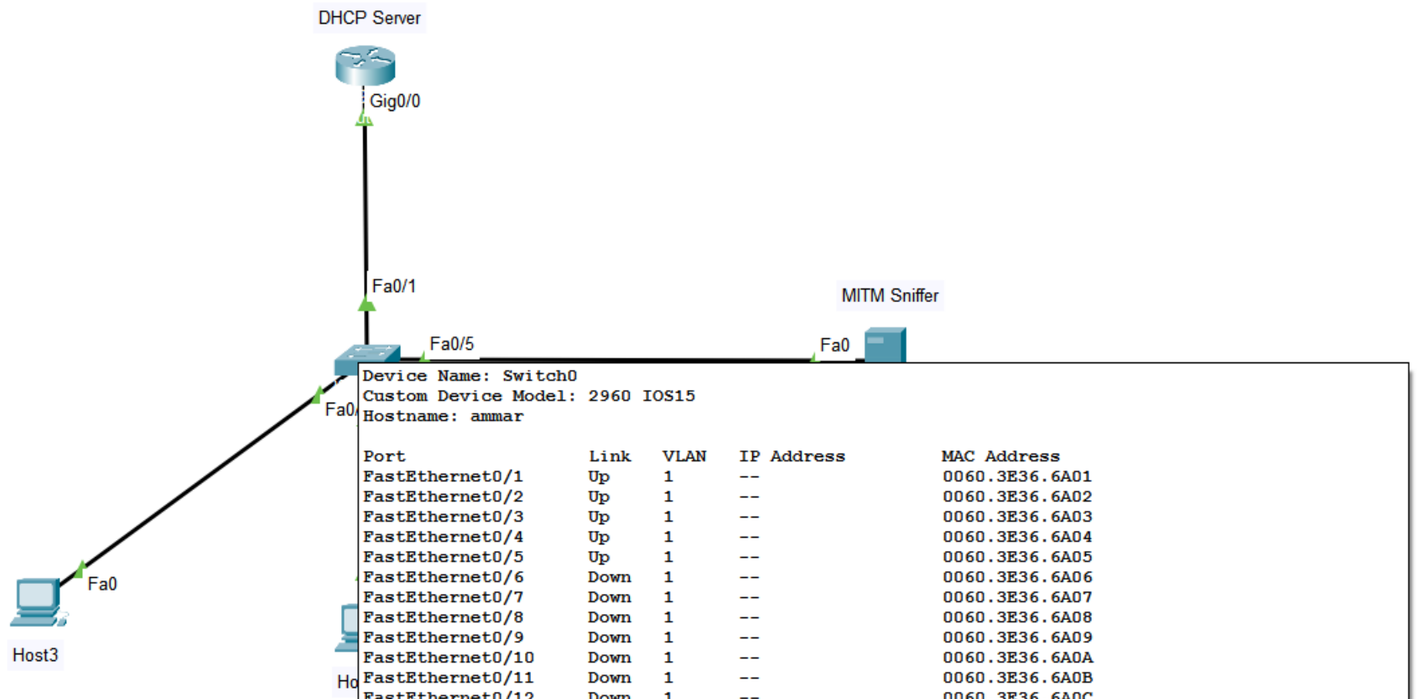

PC1
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::210:11FF:FE40:4A8B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC0
Desktop
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
169.254.170.203
Subnet Mask
255.255.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::290:21FF:FEE5:AACB
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

Cisco Packet Tracer - C:\Users\DELL\OneDrive - Faculty Of Engineering (Zagazig University)\Desktop\Projects\Cisco Packet Tracer\saves\Ammar24 (DHCP Snooping) with Port Security.pkt

File Edit Options View Tools Extensions Window Help

Logical | Physical x: 905, y: 335 | Root | 03:08:00




```
Device Name: Switch0
Custom Device Model: 2960 IOS15
Hostname: ammar

Port                Link   VLAN   IP Address          MAC Address
FastEthernet0/1     Up     1      --                  0060.3E36.6A01
FastEthernet0/2     Up     1      --                  0060.3E36.6A02
FastEthernet0/3     Up     1      --                  0060.3E36.6A03
FastEthernet0/4     Up     1      --                  0060.3E36.6A04
FastEthernet0/5     Up     1      --                  0060.3E36.6A05
FastEthernet0/6     Down   1      --                  0060.3E36.6A06
FastEthernet0/7     Down   1      --                  0060.3E36.6A07
FastEthernet0/8     Down   1      --                  0060.3E36.6A08
FastEthernet0/9     Down   1      --                  0060.3E36.6A09
FastEthernet0/10    Down   1      --                  0060.3E36.6A0A
FastEthernet0/11    Down   1      --                  0060.3E36.6A0B
FastEthernet0/12    Down   1      --                  0060.3E36.6A0C
FastEthernet0/13    Down   1      --                  0060.3E36.6A0D
FastEthernet0/14    Down   1      --                  0060.3E36.6A0E
FastEthernet0/15    Down   1      --                  0060.3E36.6A0F
FastEthernet0/16    Down   1      --                  0060.3E36.6A10
FastEthernet0/17    Down   1      --                  0060.3E36.6A11
FastEthernet0/18    Down   1      --                  0060.3E36.6A12
FastEthernet0/19    Down   1      --                  0060.3E36.6A13
FastEthernet0/20    Down   1      --                  0060.3E36.6A14
FastEthernet0/21    Down   1      --                  0060.3E36.6A15
FastEthernet0/22    Down   1      --                  0060.3E36.6A16
FastEthernet0/23    Down   1      --                  0060.3E36.6A17
FastEthernet0/24    Down   1      --                  0060.3E36.6A18
GigabitEthernet0/1  Down   1      --                  0060.3E36.6A19
GigabitEthernet0/2  Down   1      --                  0060.3E36.6A1A
Vlan1               Down   1      <not set>           000D.BDCC.B648

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch0
```

Time: 01:41:50

2960 PT-Switch PT-Empty 3560 24PS 3650 24PS IE 2000 IE-3400 IE-9320 PT-Bridge 2950-24 2950T

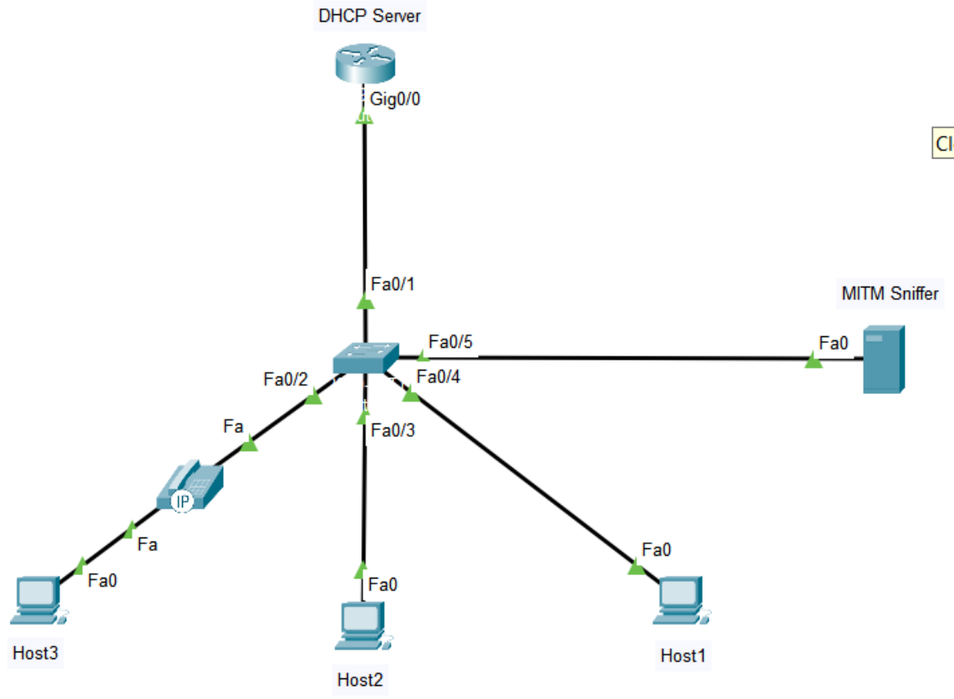
Scenario 0 | New | Delete | Toggle PDU List Windo

Realtime | Simulation

Cisco Packet Tracer - C:\Users\DELL\OneDrive - Faculty Of Engineering (Zagazig University)\Desktop\Projects\Cisco Packet Tracer\saves\Ammar24 (DHCP Snooping) with Port Security.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1335, y: 159 Root 03:26:30



Time: 00:06:47 Realtime Simulation

Scenario 0 New Delete Toggle PDU List Window

Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete

Switch0

CLI

```
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

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Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


ammar>en
Password:
ammar#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ammar(config)#int ran f0/1 -4
ammar(config-if-range)#swi po vio ?
  protect   Security violation protect mode
  restrict  Security violation restrict mode
  shutdown  Security violation shutdown mode
ammar(config-if-range)#swi po vio res
ammar(config-if-range)#int ran f0/2
ammar(config-if-range)#swi po max 2
ammar(config-if-range)#
%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
```

Copy Paste

Top

Switch0

CLI

```
Model number                    : WS-C2960-24TT-L
System serial number            : FOC1010X104
Top Assembly Part Number        : 800-27221-02
Top Assembly Revision Number    : A0
Version ID                      : V02
CLEI Code Number                : COM3L00BRA
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-LANBASEK9-M

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Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

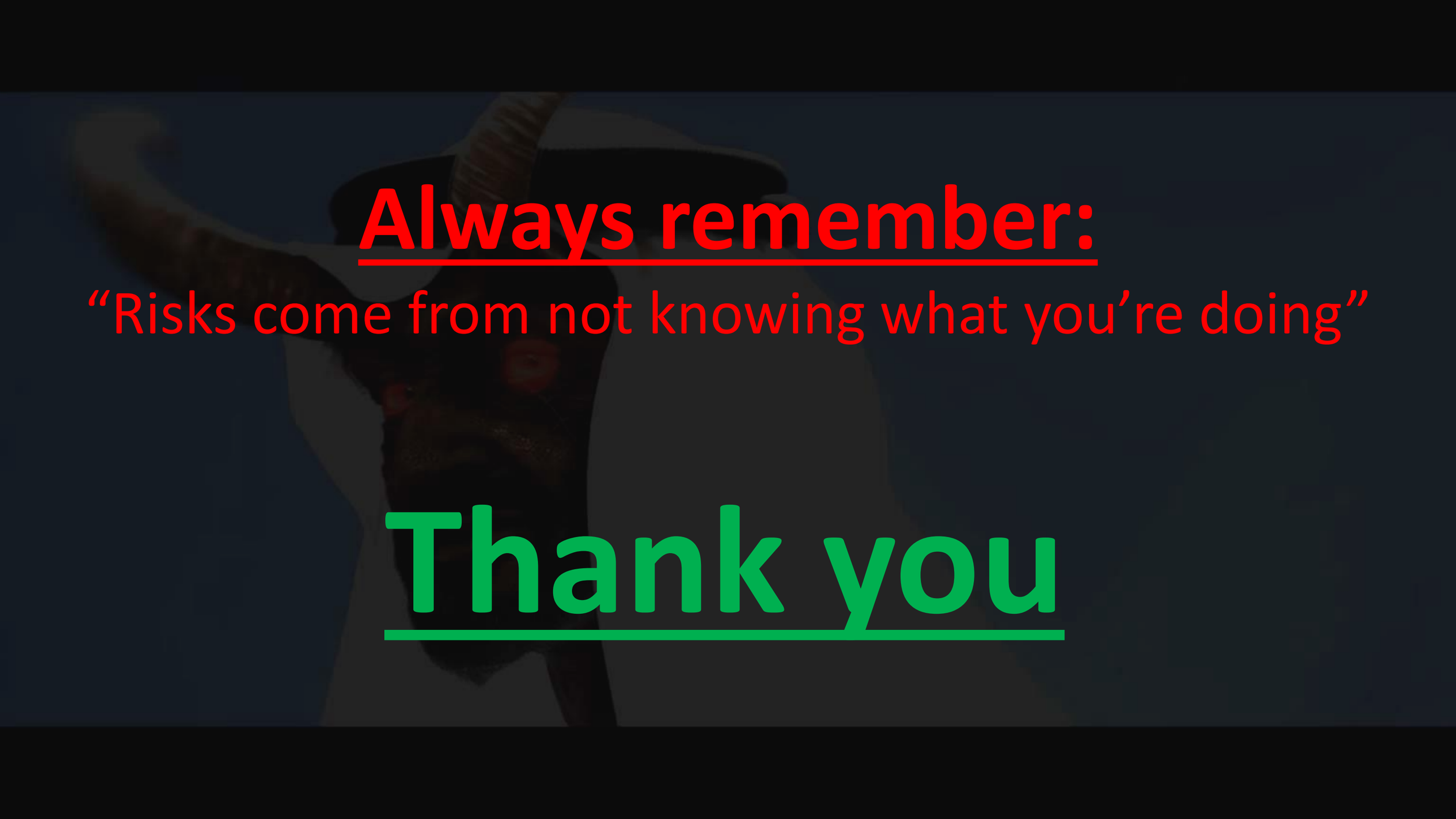
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


ammar>en
Password:
ammar#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ammar(config)#int ran f0/1 -4
ammar(config-if-range)#swi po vio ?
  protect   Security violation protect mode
  restrict  Security violation restrict mode
  shutdown  Security violation shutdown mode
ammar(config-if-range)#swi po vio res
ammar(config-if-range)#
```

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Top



Always remember:

"Risks come from not knowing what you're doing"

Thank you